

Q-Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface.
Hint: You can map hostnames to local IPs using the DNS service or by editing the hosts file for a quick test.

To map a custom hostname (musicplayer.local) to a local IP address using the Windows hosts file and verify that the hostname opens a local web page.

Step 1: Open the Hosts File

1. Open **Notepad** as **Administrator**.
2. Click **File** → **Open**.
3. Navigate to:

C:\Windows\System32\drivers\etc\

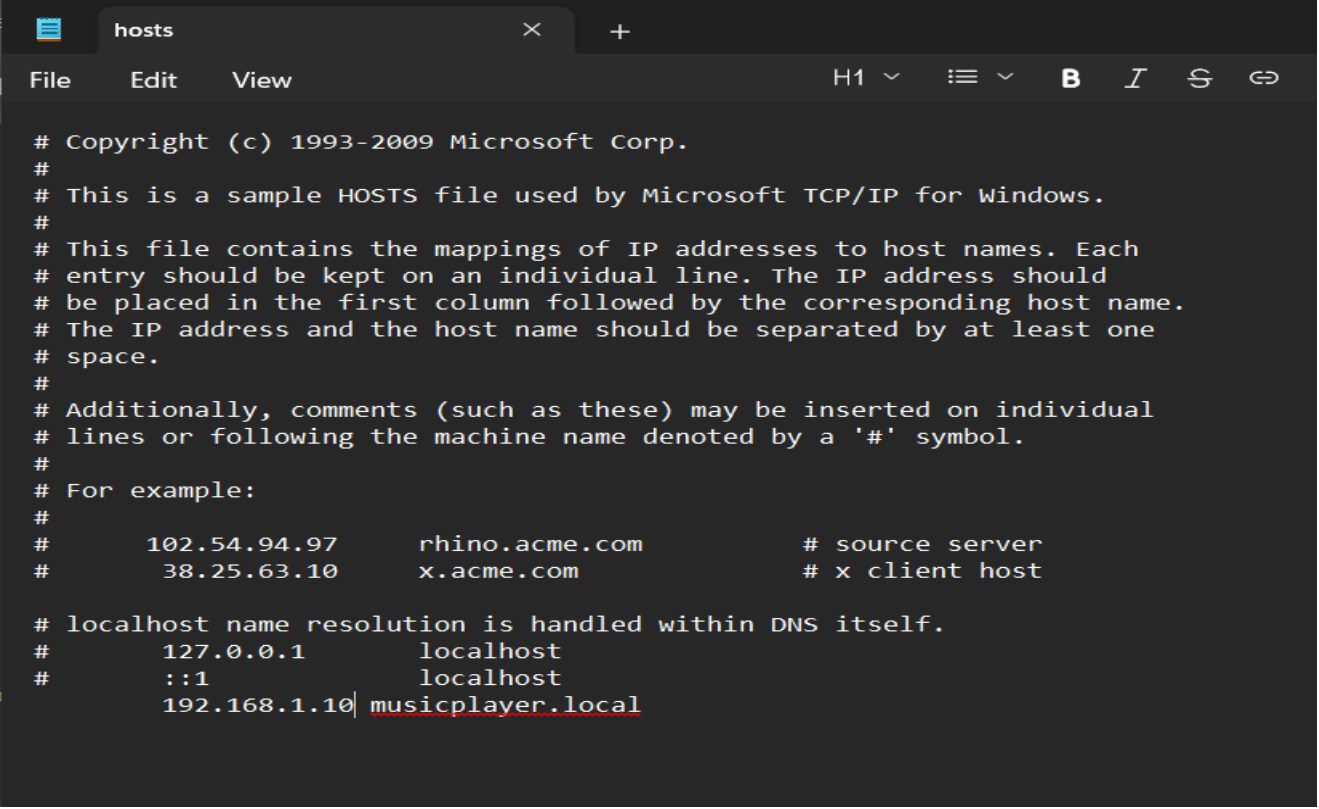
4. Change the file type from **Text Documents (*.txt)** to **All Files (.)**.
5. Open the **hosts** file.

Step 2: Add the Hostname Mapping

Add the following line at the end of the file:

192.168.1.10 musicplayer.local

Save the file.

A screenshot of a Windows Notepad window titled 'hosts'. The window has a menu bar with 'File', 'Edit', and 'View'. The text inside the Notepad is as follows:

```
# Copyright (c) 1993-2009 Microsoft Corp.
#
# This is a sample HOSTS file used by Microsoft TCP/IP for Windows.
#
# This file contains the mappings of IP addresses to host names. Each
# entry should be kept on an individual line. The IP address should
# be placed in the first column followed by the corresponding host name.
# The IP address and the host name should be separated by at least one
# space.
#
# Additionally, comments (such as these) may be inserted on individual
# lines or following the machine name denoted by a '#' symbol.
#
# For example:
#
#       102.54.94.97       rhino.acme.com           # source server
#       38.25.63.10      x.acme.com               # x client host
#
# localhost name resolution is handled within DNS itself.
#       127.0.0.1        localhost
#       ::1              localhost
#       192.168.1.10| musicplayer.local
```

The last line, '192.168.1.10| musicplayer.local', is highlighted with a red selection box.

Step 3: Verify the Configuration

- 1. Open **Command Prompt**.
- 2. Run the following command:

```
ping musicplayer.local
```

- 3. Confirm that the hostname resolves to **192.168.1.10**.

Step 4: Test in a Browser

- 1. Open a web browser.
- 2. Enter:

```
http://musicplayer.local
```

- 3. Verify that the local web page opens successfully.

Observations

Parameter	Value
Operating System	Windows
Local Web Server IP	192.168.1.10
Hostname	musicplayer.local
Hosts File Modified	Yes
Ping Test	Successful
Browser Test	Web page loaded successfully

The Windows **hosts** file was successfully configured to map the hostname **musicplayer.local** to the local IP address **192.168.1.10**. The system resolved the hostname correctly, and the local web page was successfully accessed using the custom hostname instead of the IP address.